Why Self-Arbitration Looks Like Guessing: A Diagnostic Study of Small Language Models

Manh Khang Tran
Independent researcher
https://github.com/KhangTranManh/Self-Correction-Model

Preprint, October 2026
DOI: 10.5281/zenodo.23541462

Abstract

When a 7B language model solves a problem twice, gets two different answers, and is asked which of its own solutions is right, it looks close to guessing: judged in a single presentation order, it picks the correct solution 45–64% of the time. We ask whether this reflects a lack of skill or the way it is measured. On self-generated solution pairs with deterministic labels (four models from three families, four math datasets, both presentation orders), a skill index that cancels pure position bias is clearly positive in every cell we measured: 4/4 preregistered endpoint cells (Llama-3.1-8B, Mistral-7B × ASDiv, MATH levels 1–3) and 16/16 descriptive cells. Decomposing the single-order shortfall shows that position bias costs only 0–2.5 accuracy points (at most 1.9 under a probit link); most of the gap comes from the judge not making a valid choice between the two solutions — writing its choice in an unreadable format, giving a third answer, or being cut off at the token limit while looping. Only 51–90% of single-order judgments are valid choices. Part of this is an answer-format artefact: re-reading the judge's final line raises the untrained Qwen2.5-7B-Instruct from 45–64% to 72–88%. For Qwen the split between position bias and invalid choices is not clean: it fails to choose more often when the correct solution is shown second, so part of its position preference appears as invalid choices. Averaging verdict probabilities over both orders, the debiasing method of Ren et al. [2023], adds little (1/4 preregistered cells reach $\geq$3 points). Hidden states carry a right/wrong signal in all four models, but beat simple surface features consistently only for the self-correction model. At the system level, using the judge's verdict loses to majority voting over three samples in 15/16 cells; falling back to a third sample when the two orders disagree only ties three-sample voting, and every two-candidate strategy loses to voting over five.

1 Introduction

Self-correction involves three steps: noticing that an answer may be wrong, deciding which of several candidate answers is right, and repairing the answer. Prior work finds that small and medium models struggle to self-correct reasoning without external feedback [Huang et al., 2023], and that the bottleneck is locating errors rather than fixing them [Tyen et al., 2024]. We study the middle step, *arbitration*: given two of its own solutions with different final answers, can the model tell which one is right?

Arbitration is usually scored in a single presentation order. Position bias is well documented in LLM-as-a-judge settings [Zheng et al., 2023, Wang et al., 2024, Shi et al., 2024], but it is rarely separated from skill when a model judges its own solutions. We ask whether the conclusion "the model is guessing" reflects a lack of skill or the measurement.

Contributions. (i) A decomposition of single-order arbitration accuracy into skill, position bias and *invalid choices*, applied to self-generated solution pairs with deterministic labels. (ii) A preregistered test on two model families and two fresh datasets showing that the skill is real and general (E1, 4/4 cells), while order-averaging adds little (E2, 1/4 cells). (iii) Evidence that the dominant cost of single-order arbitration is not position bias but the judge failing to make a valid choice (format errors, third answers, truncation while looping), with a seven-way classification of judgment outcomes and bounds that do not depend on the model of the decomposition. (iv) A representation-level analysis showing a right/wrong signal in hidden states that is only partly distinct from surface cues, and a practical comparison with majority voting.

2 Related work

Self-correction. Intrinsic self-correction often fails to improve reasoning [Huang et al., 2023]; models find errors poorly but fix them when told where they are [Tyen et al., 2024]. Training methods such as SCoRe [Kumar et al., 2025] and RISE [Qu et al., 2024] teach multi-turn self-correction. We isolate arbitration between two of the model's own solutions.

LLM judges and position bias. LLM judges prefer answers by position [Zheng et al., 2023, Wang et al., 2024]; the size of the bias depends on the quality gap between candidates [Shi et al., 2024]. JudgeBench [Tan et al., 2025] scores judges on pairs with objective labels and finds strong judges near chance; J1 [Whitehouse et al., 2026] and permutation-aware RL [Zheng et al., 2026] train judges for order consistency. Self-preference is a competing explanation when a model judges its own text [Panickssery et al., 2024].

Selecting among own samples. Universal self-consistency lets the model pick the most consistent answer [Chen et al., 2023]. Ren et al. [2023] report that averaging letter probabilities over shuffled orders barely changes selection accuracy (+0.5–0.6 points); we measure the same quantity in our setting.

Probing for correctness. Hidden states encode the truth of statements [Burns et al., 2023, Azaria and Mitchell, 2023, Marks and Tegmark, 2023] and models can encode the right answer while producing a wrong one, with probes that transfer poorly across datasets [Orgad et al., 2025]; P(True)-style self-evaluation is calibrated in large models [Kadavath et al., 2022].

Positioning. We do not propose order swapping (it is standard). We use it to separate bias from skill on self-generated pairs, show that the remaining shortfall is mostly invalid choices, and connect the output-level picture to hidden states under a preregistered protocol.

3 Setup

Models. Self_Correction_v1, a Qwen2.5-7B-Instruct fine-tuned for self-correction (one SFT stage on verified correction traces); its untrained parent Qwen2.5-7B-Instruct; Llama-3.1-8B-Instruct; and Mistral-7B-Instruct-v0.3. Two LoRA judges trained by DPO on order-swapped data (p12, p13b) are used only in the single-model analysis (§4.6).

Data. GSM8K test problems 750–1318 [Cobbe et al., 2021] (561; 519 after removing 42 that appeared in an earlier evaluation-only run), SVAMP (960) [Patel et al., 2021], and two fresh datasets frozen before any generation: ASDiv (822) [Miao et al., 2020] and MATH levels 1–3 with numeric answers (785) [Hendrycks et al., 2021]. No problem overlaps earlier phases or training data.

Pairs. Each model samples five solutions per problem (temperature 0.7); the first 768 tokens are used. Pairs (s_1, s_2) and (s_1, s_3) whose final answers differ are judged in both orders ("ab": s_1 shown as A). The main analysis uses *one-right* pairs, where exactly one solution is correct.

Judges. Each model judges its own pairs, greedily, with a 768-token cap. `base`: write a check, then a free final answer, mapped to A, B, or neither. `base_c`: must end with "Verdict: Solution A/B"; log-probabilities of A and B are recorded at the verdict token.

Labels. A deterministic SymPy verifier; no LLM labels. The project parser is primary; a "rescued" reading (first number on the last answer line when the parser yields no number) is a sensitivity check.

Measures. Single-order accuracy; *skill index* $= P(\text{pick A} \mid \text{A right}) - P(\text{pick A} \mid \text{B right})$, which a pure position bias cancels; position-2 rate; tie-break score (consistent right 1, consistent wrong 0, otherwise 0.5); consistent accuracy and coverage. Confidence intervals are problem-clustered bootstrap (10,000 resamples for endpoints).

Preregistration. Endpoints were fixed before generation and opened once: **E1** skill index of the `base` judge with CI above 0; **E2** gain of order-averaged verdict log-probabilities (sign of $m_{ab} - m_{ba}$, $m = \log P(A) - \log P(B)$) over single-order accuracy of the same log-probabilities, $\geq$3 points with CI above 0. A claim is "general" if it holds in $\geq 3/4$ endpoint cells. All other cells and analyses are descriptive or exploratory and are labeled as such.

4 Results

4.1 Single-order arbitration looks like guessing, but the skill is real

Table 1 reports the `base` judge in all 16 cells. Single-order accuracy is 45–64%, yet the skill index is positive with a CI above 0 in every cell. **E1 passes in 4/4 endpoint cells** (Llama ASDiv +0.321 [0.230, 0.413], MATH +0.362 [0.291, 0.428]; Mistral ASDiv +0.335 [0.269, 0.400], MATH +0.344 [0.261, 0.431]). The direction of the position bias is a stable property of each model across all four datasets: the three untrained models favor the first position (position-2 rate 34–45%), the self-correction model favors the second (52–56%), consistent with — but not proof of — its SFT data: in all 513 training rows the first attempt is marked wrong by the verifier and the later answer is the verified correction.

4.2 Where the single-order shortfall comes from

Seven outcomes of a judgment (exploratory). We assign each single-order judgment to exactly one of seven classes, checked in order: (1) a valid choice by the project parser; (2) a format error — no valid choice, but the first number on the final answer line matches exactly one solution (the *rescued* reading); (3) cut at the token cap while looping; (4) cut at the cap otherwise; (5)

Table 1: `base` judge, project parser, one-right pairs. Endpoint cells in bold. Skill index with 95% CI. Pos. 2 = share of valid choices for the second position; Invalid = share of judgments with no valid choice (%).

Model	Dataset	Pairs	Single	Skill index [CI]	Pos. 2	Invalid	Tie-break
Llama-3.1-8B	**ASDiv**	274	57.3	0.321 [0.230, 0.413]	44.3	19.3	0.646
	MATH 1–3	434	53.0	0.362 [0.291, 0.428]	43.9	26.7	0.628
	GSM8K	264	50.2	0.307 [0.211, 0.403]	40.9	29.2	0.604
	SVAMP	389	56.4	0.357 [0.278, 0.434]	41.7	22.1	0.647
Mistral-7B	**ASDiv**	558	51.3	0.335 [0.269, 0.400]	45.4	29.2	0.616
	MATH 1–3	331	50.5	0.344 [0.261, 0.431]	41.9	31.4	0.609
	GSM8K	411	47.4	0.255 [0.180, 0.332]	41.5	28.7	0.575
	SVAMP	681	47.4	0.254 [0.194, 0.313]	44.3	29.2	0.584
Self_Correction	ASDiv	311	55.9	0.222 [0.127, 0.313]	52.2	10.9	0.603
	MATH 1–3	452	49.0	0.155 [0.078, 0.229]	55.7	19.6	0.573
	GSM8K	277	52.2	0.191 [0.095, 0.283]	55.4	13.7	0.590
	SVAMP	371	56.3	0.237 [0.149, 0.324]	54.8	10.0	0.598
Qwen2.5-7B-Instr.	ASDiv	256	63.5	0.609 [0.532, 0.682]	36.0	25.6	0.701
	MATH 1–3	380	45.1	0.466 [0.405, 0.526]	41.3	49.1	0.609
	GSM8K	241	56.4	0.485 [0.399, 0.567]	41.9	31.1	0.651
	SVAMP	358	63.1	0.575 [0.514, 0.638]	33.9	22.2	0.679

re-solved the problem from scratch and reached a third answer; (6) another third answer; (7) no number at all. The rules were written down before the analysis was run. Figure 1 shows the `base` judge: only 51–90% of judgments are valid choices. The failure profile is model-specific. Qwen2.5-7B-Instruct mostly writes its choice as a sentence (format errors, 16–26% of judgments). Mistral mostly gives a third answer or re-solves. Llama is often cut off while looping, especially on MATH. With the constrained `base_c` prompt, 79–99% of judgments are valid.

Decomposing the gap. Let $R = \sigma(\delta)$ be the accuracy of a judge with the estimated skill δ, no position bias and no invalid choices (δ and bias β fitted on valid judgments only). We split the gap between R and single-order accuracy into four factors with Shapley values over the 24 removal orders: position bias, format errors (class 2), truncation (3–4), and no choice despite a finished output (5–7). Table 2 gives the split. In the 16 grid cells the gap is 6.4–43.3 points; position bias accounts for 0.0–2.5 of it, and the three invalid-choice factors for the rest: format errors 0.2–22.8 (largest for Qwen), truncation 0.5–13.9 (largest for Llama and for Qwen on MATH), and no choice 2.6–16.4 (largest for Mistral). With `base_c` the gap shrinks to 2.1–13.6 points.

Sensitivity. The split does not hinge on the logistic model. (i) In a model-free view, single-order accuracy equals $(1-\pi)\,\mathrm{acc}_{AB}$, with π the invalid share and acc_{AB} the accuracy of valid judgments; this identity holds to machine precision in every cell. The Shapley share of invalid choices always lies above the lower bound 0.5π (invalid judgments would have been coin flips). It equals the upper bound $\pi\,\mathrm{acc}_{AB}$ to within 0.1 points (invalid judgments would have been as accurate as valid ones). (ii) Replacing the logistic link with a probit changes the bias share by at most 0.6 points; the largest bias share under probit is 1.9 points, inside the preregistered 3-point criterion. (iii) The separation is not fully clean for Qwen. Its invalid share depends on where the correct solution sits: it is 7–18 points higher when the correct solution is second, with CIs excluding 0 in 5/5 Qwen cells and in Llama MATH. So for Qwen part of the position preference shows up as failing to choose, not as a

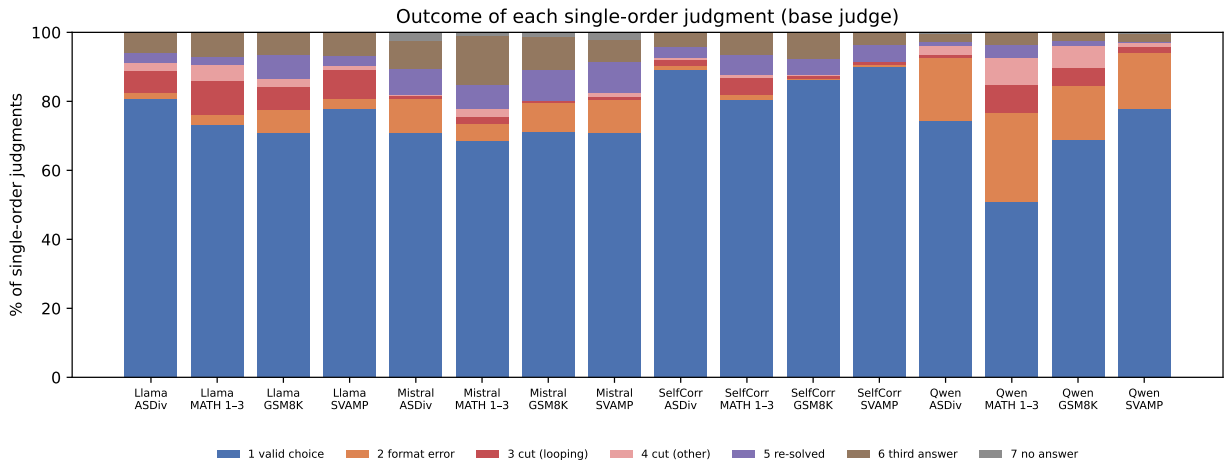


Figure 1: Outcome of each single-order judgment (`base` judge, one-right pairs, both orders), 16 grid cells. Exploratory classification; rules fixed before the analysis.

wrong choice.

The token cap is not the cause: re-judging Llama's endpoint pairs with a 1536-token cap lowers the share of truncated judgments (18.1% → 12.8% on ASDiv, 31.9% → 15.8% on MATH) but leaves the gap unchanged (14.3 → 14.5 points on ASDiv, 19.4 → 19.6 on MATH); the extra tokens end in an output that still names neither solution, consistent with most truncated judgments being loops.

4.3 Order averaging adds little (comparison with Ren et al.)

Table 3 compares order averaging on the scale of Ren et al. [2023]. **E2 passes in 1/4 endpoint cells** (Llama MATH +5.0 [0.9, 9.3]; others +0.6, +0.7, +3.3 with CIs including 0). The bias-removal part of the gain is near 0 in every endpoint cell although the log-odds bias toward position A is large (1.3–2.4); the preregistered prediction that this part grows with |bias| is not supported ($\rho = -0.20$ over 4 cells, -0.27 over 16). In descriptive cells the gain is heterogeneous (0–11 points; 8/16 cells $\geq$3 points with CI above 0), and mostly comes from reading two orders rather than removing a bias. Note that tie-break minus single-order accuracy is *not* a debiasing gain: per pair it equals half the indicator that no order picks the right solution and at least one invents an answer (Appendix E).

4.4 Hidden states

A linear probe (standardized features, L2 logistic regression, $C = 0.01$, layer chosen by grouped CV on the training set) reads which solution is right from the last token of the judge prompt in all four models (cross-dataset AUC 0.66–0.90). The probe beats a surface-feature baseline (length, step count, answer magnitude) in every train/test direction only for Self_Correction (e.g. 0.814 vs 0.690 for SVAMP→GSM8K); for Llama only on the endpoint datasets; for Mistral and Qwen surface features are stronger (Qwen 0.91–0.95, likely the answer-format artefact). A pointwise probe that scores one solution at a time is stronger than the pairwise probe (AUC 0.69–0.97). No probe is order-invariant: its choice flips with the order in 22–72% of pairs. Log-probability of the solution is a weak selector (48–69%).

Table 2: Four-factor Shapley split of the gap $R-$single-order accuracy (`base` judge, points). "Valid" = share of judgments with a valid choice. Gap with 95% problem-clustered bootstrap CI. Exploratory.

Model	Dataset	Valid	Single	Gap [CI]	Bias	Format	Cut	No choice
Llama-3.1-8B	ASDiv	80.7	57.3	14.3 [11.5, 17.5]	0.4	1.3	6.2	6.3
	MATH 1–3	73.3	53.0	19.4 [16.9, 22.3]	0.2	2.1	10.3	6.9
	GSM8K	70.8	50.2	21.3 [17.8, 25.5]	0.6	4.7	6.3	9.7
	SVAMP	77.9	56.4	16.7 [13.9, 19.6]	0.6	2.1	6.8	7.1
Mistral-7B	ASDiv	70.8	51.3	21.3 [18.8, 23.9]	0.1	7.3	0.8	13.0
	MATH 1–3	68.6	50.5	23.5 [20.1, 27.0]	0.5	3.6	3.1	16.4
	GSM8K	71.3	47.4	19.4 [16.7, 22.2]	0.3	5.4	0.5	13.1
	SVAMP	70.8	47.4	19.6 [17.6, 22.0]	0.1	6.4	1.3	11.8
Self_Correction	ASDiv	89.1	55.9	6.9 [5.2, 8.6]	0.0	0.8	1.4	4.6
	MATH 1–3	80.4	49.0	11.9 [10.0, 14.1]	0.1	0.9	3.5	7.4
	GSM8K	86.3	52.2	8.5 [6.9, 10.5]	0.2	0.2	0.7	7.4
	SVAMP	90.0	56.3	6.4 [5.0, 7.9]	0.2	0.4	0.5	5.4
Qwen2.5-7B-Instr.	ASDiv	74.4	63.5	23.7 [20.2, 28.5]	2.3	15.2	3.0	3.3
	MATH 1–3	50.9	45.1	43.3 [39.1, 48.2]	0.2	22.8	13.9	6.5
	GSM8K	68.9	56.4	25.8 [22.0, 30.0]	0.5	12.5	9.7	3.0
	SVAMP	77.8	63.1	20.0 [17.1, 23.0]	2.5	12.8	2.1	2.6

Table 3: Single-order vs order-averaged verdict accuracy (`base_c` log-probabilities), same scale as Ren et al. [2023]. Gain with 95% CI.

Study / model	Dataset	Single → averaged	Gain [CI]	Bias removal
Ren et al., PaLM-2 L	TruthfulQA	58.3 → 58.9	+0.6	—
Ren et al., PaLM-2 L	TL;DR	70.2 → 70.7	+0.5	—
Llama-3.1-8B	ASDiv	61.8 → 62.4	+0.6 [−3.7, 4.8]	−0.8
	MATH 1–3	62.4 → 67.4	**+5.0** [0.9, 9.3]	+1.4
Mistral-7B	ASDiv	71.7 → 72.4	+0.7 [−2.3, 3.6]	−0.3
	MATH 1–3	62.7 → 66.0	+3.3 [−0.7, 7.3]	+1.2
Self_Correction	GSM8K (report pairs)	56.7 → 64.9	+8.2 [3.6, 12.8]	+2.2
Qwen2.5-7B-Instr.	ASDiv / MATH / GSM8K / SVAMP	55–59 → 57–63	−0.2 to +6.0	−0.8 to +2.7

Probe-weighted voting out of sample (exploratory). We trained a pointwise probe on the answer-site state of s_1–s_3 on one dataset and used it on another. Each answer cluster was weighted by the summed probe probability, and the heaviest cluster was chosen. Three train→test pairs were used (ASDiv→MATH, MATH→ASDiv, SVAMP→GSM8K). The criterion, fixed before running, was a paired CI above 0 in at least 2/3 pairs. Weighted voting beats vote@3 by +1.7 to +3.4 points for Llama (2/3 pairs) and by +3.2 to +4.4 for Mistral (3/3), so it is confirmed out of sample for these two models. It is not confirmed for Self_Correction (1/3; +0.6 to +3.4) or for Qwen (0/3; −1.1 to +1.6), whose vote@3 is already high. The test AUC is 0.77–0.90 in all models. The gains are smaller than the gap between vote@3 and vote@5, and we cannot test a probe-weighted vote@5 because no hidden states were saved for s_4–s_5.

4.5 Practical value

Table 4 and Figure 2 compare strategies by accuracy per problem and mean number of model calls. The judge is called only when $s_1 \neq s_2$ (22–87% of problems), in both orders, so it costs $2 + 2P(s_1 \neq s_2) = 2.4$–$3.7$ calls. Taking the judge's consistent choice, and otherwise a random

Table 4: Accuracy (%) per problem, project parser. Judge = consistent choice over both orders, otherwise random; self-check = otherwise vote@3. Ceiling = s_1 or s_2 correct. Paired bootstrap CI. Exploratory.

Model	Dataset	Keep s_1	Judge	Self-check	vote@3	vote@5	Ceiling	Judge − vote@3 [CI]
Llama-3.1-8B	ASDiv	79.6	81.8	83.8	83.7	87.1	88.0	−1.9 [−3.5, −0.4]
	MATH 1–3	44.2	47.4	51.5	50.8	57.3	57.8	−3.4 [−5.5, −1.4]
	GSM8K	74.3	74.2	78.3	78.3	82.4	83.6	−4.0 [−6.1, −1.9]
	SVAMP	75.7	78.6	81.1	80.8	84.9	86.5	−2.2 [−3.8, −0.6]
Mistral-7B	ASDiv	54.6	61.6	68.1	68.5	75.3	74.9	−6.9 [−9.1, −4.7]
	MATH 1–3	15.8	22.1	25.0	23.8	28.8	30.1	−1.7 [−3.8, 0.3]
	GSM8K	43.1	48.3	54.9	55.6	65.1	64.5	−7.3 [−10.3, −4.4]
	SVAMP	51.8	58.6	65.4	66.2	75.0	74.0	−7.6 [−9.9, −5.5]
Self_Correction	ASDiv	79.6	81.1	83.5	85.6	88.3	88.8	−4.5 [−6.2, −2.8]
	MATH 1–3	50.2	50.8	54.0	56.4	61.4	62.4	−5.7 [−8.0, −3.4]
	GSM8K	65.8	66.5	69.5	71.1	76.8	76.8	−4.6 [−7.0, −2.2]
	SVAMP	82.7	83.5	85.4	87.0	90.3	91.6	−3.4 [−5.1, −1.8]
Qwen2.5-7B-Instr.	ASDiv	85.6	87.5	91.6	91.0	92.7	92.5	−3.5 [−4.7, −2.2]
	MATH 1–3	55.9	57.0	66.4	66.6	71.3	66.8	−9.6 [−11.4, −7.9]
	GSM8K	81.1	81.3	85.4	84.7	89.1	88.6	−3.4 [−5.2, −1.5]
	SVAMP	85.0	86.3	90.4	90.1	92.7	92.6	−3.8 [−5.2, −2.4]

one, *loses* to three-sample voting by 1.7–9.6 points, with the paired CI below 0 in 15/16 cells (13/16 under the rescued reading). Self-check, which falls back to a third sample when the two orders disagree, is within −2.4 to +1.1 points of vote@3 (CI above 0 in no cell) at 2.6–4.4 calls. Agreement-gated voting (stop if $s_1 = s_2$) gives exactly the vote@3 answer at 2.2–2.9 calls, and is the cheapest way to get it. Five-sample voting beats every two-candidate strategy in 15–16/16 cells. Even a perfect arbiter could not change this much: 20–76% of disagreements have both solutions wrong, so the ceiling (correct if s_1 or s_2 is correct) is at most 1.6 points above vote@5, and not above it in 8/16 cells.

4.6 A closer look at the self-correction model

On the original GSM8K pairs the untrained judge has skill index +0.215 [0.128, 0.304] and tie-break 0.601; order-swapped DPO raises tie-break by 5.2 points (Holm $p = 0.016$) and cuts invented answers from 16.7% to 2.1%, while a probe trained on the same states is unchanged across all four judges (65–69%): training changes how the signal is expressed, not the signal. The Phase-14 pipeline reproduces every number of this analysis exactly from the saved outputs, and a fresh sample reproduces it within its CI (skill 0.191).

5 Discussion

What makes arbitration look like guessing. Position bias is real and model-specific, but costs little accuracy because a moderate symmetric bias reduces accuracy only to second order. The large cost is the judge failing to commit to one of the two solutions. For instruction models this is partly a reading problem (answers written as sentences); for the others it is invented third answers or loops cut by the token limit.

A weak point: Qwen. The decomposition assumes that how often a judge fails to choose does not depend on where the correct solution is shown. This holds in 15 of 21 cells but not for Qwen:

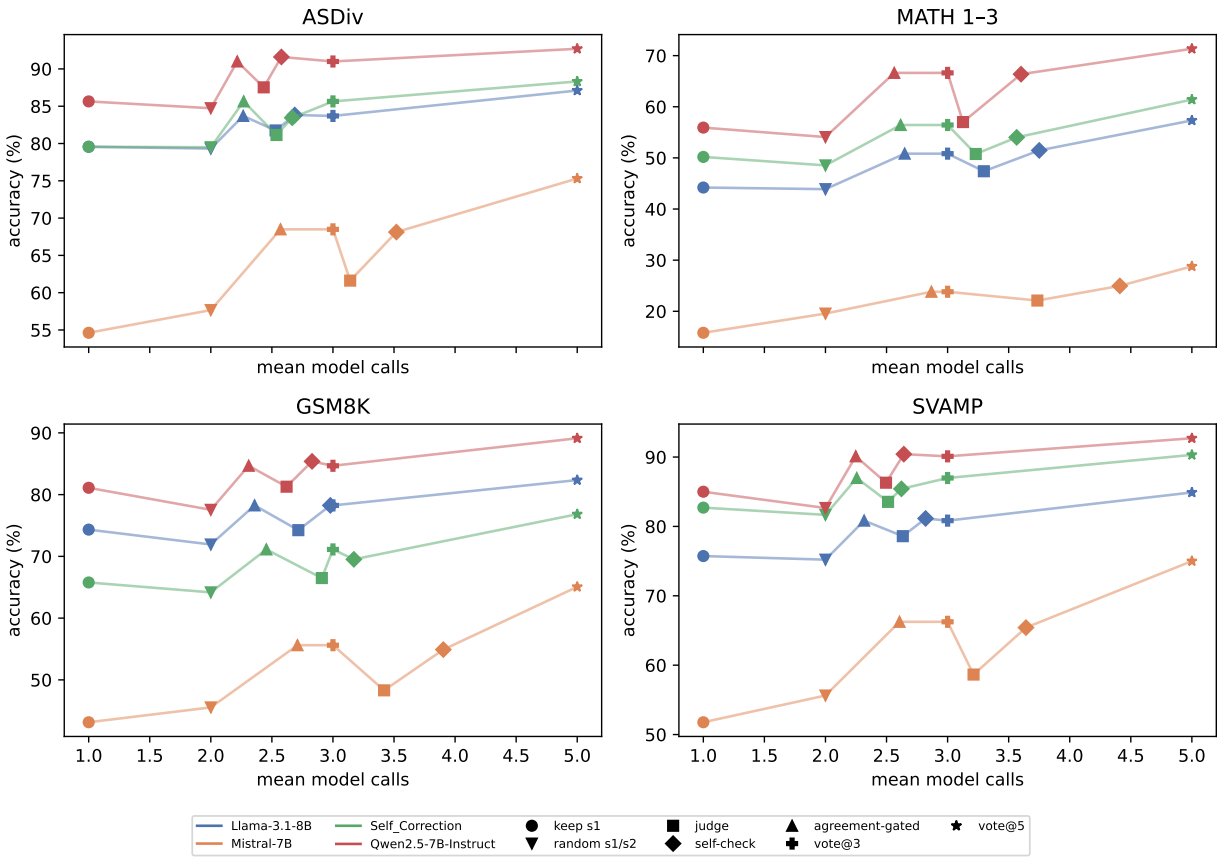


Figure 2: Accuracy vs mean model calls per problem, one panel per dataset; colors are models, markers are strategies.

its invalid share is 7–18 points higher when the correct solution is second (CIs excluding 0 in all 5 Qwen cells, and in Llama MATH). For these cells part of the position preference is expressed as a failure to choose rather than a wrong choice, so the bias share in Table 2 is a lower bound and the invalid-choice share an upper bound. The conclusion that bias costs little is weakest for Qwen.

Parent vs fine-tuned model (open question). The untrained parent has a 2–3× larger skill index than the self-correction model on all four datasets, and the gap widens under the rescued reading. Each model judges its own pairs, which differ in difficulty, so we do not claim that self-correction SFT reduces arbitration skill; testing this needs both models to judge the same pairs.

Recommendation. When a model arbitrates its own solutions, swap-test, report single-order accuracy together with the skill index and tie-break, and report how often the judge fails to make a valid choice.

6 Limitations

Endpoints cover two models × two datasets; the other cells are descriptive and were generated after the endpoints were opened. All prompts are in Vietnamese; Mistral is a weak solver here and often answers in English. The project parser misreads answers written as sentences. Judges

decode greedily without a repetition penalty, which affects looping. Log-probabilities exist only for the constrained judge and cover 50–94% of pairs. For Qwen the invalid share depends on the position of the correct solution, so the separation of position bias from invalid choices is not fully clean for that model. The check of the rescued reading was done by the analysing assistant, not by an independent annotator. One run per cell. Llama weights come from a byte-identical public copy of Meta's release. Most analyses beyond E1/E2 are post hoc and need confirmation on new data.

7 Conclusion

Measured in one order, a 7B model arbitrating between its own conflicting solutions looks like it is guessing. Decomposed, the skill is real and general across model families and datasets; the shortfall comes mostly from the judge not making a valid choice, and only a little from position bias. Constraining the output format removes most invalid choices. Order averaging adds little, and at the system level self-arbitration does not beat majority voting.

Data and code availability. All numbers in this paper are recomputed on CPU from saved model outputs; generation used one rented GPU. Analysis scripts, saved outputs, the data catalog with SHA-256 hashes, and the preregistration with every amendment are in the project repository, `https://github.com/KhangTranManh/Self-Correction-Model` (archived version: doi:10.5281/zenodo.23541462).

References

Amos Azaria and Tom Mitchell. The internal state of an LLM knows when it's lying. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023. arXiv:2304.13734.

Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. Discovering latent knowledge in language models without supervision. In *International Conference on Learning Representations*, 2023. arXiv:2212.03827.

Xinyun Chen, Renat Aksitov, Uri Alon, et al. Universal self-consistency for large language model generation. *arXiv preprint arXiv:2311.17311*, 2023.

Karl Cobbe et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Dan Hendrycks et al. Measuring mathematical problem solving with the MATH dataset. In *NeurIPS Datasets and Benchmarks*, 2021.

Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. Large language models cannot self-correct reasoning yet. *arXiv preprint arXiv:2310.01798*, 2023.

Saurav Kadavath et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

Aviral Kumar et al. Training language models to self-correct via reinforcement learning. In *International Conference on Learning Representations*, 2025. arXiv:2409.12917.

Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. *arXiv preprint arXiv:2310.06824*, 2023.

Shen-yun Miao, Chao-Chun Liang, and Keh-Yih Su. A diverse corpus for evaluating and developing English math word problem solvers. In *ACL*, 2020.

Hadas Orgad, Michael Toker, Zorik Gekhman, Roi Reichart, Idan Szpektor, Hadas Kotek, and Yonatan Belinkov. LLMs know more than they show: On the intrinsic representation of LLM hallucinations. In *International Conference on Learning Representations*, 2025. arXiv:2410.02707.

Arjun Panickssery, Samuel R. Bowman, and Shi Feng. LLM evaluators recognize and favor their own generations. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2404.13076.

Arkil Patel, Satwik Bhattamishra, and Navin Goyal. Are NLP models really able to solve simple math word problems? In *NAACL*, 2021.

Yuxiao Qu, Tianjun Zhang, Naman Garg, and Aviral Kumar. Recursive introspection: Teaching language model agents how to self-improve. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2407.18219.

Jie Ren, Yao Zhao, Tu Vu, Peter J. Liu, and Balaji Lakshminarayanan. Self-evaluation improves selective generation in large language models. *arXiv preprint arXiv:2312.09300*, 2023.

Lin Shi, Chiyu Ma, Wenhua Liang, Xingjian Diao, Weicheng Ma, and Soroush Vosoughi. Judging the judges: A systematic study of position bias in LLM-as-a-judge. *arXiv preprint arXiv:2406.07791*, 2024.

Sijun Tan et al. JudgeBench: A benchmark for evaluating LLM-based judges. In *International Conference on Learning Representations*, 2025. arXiv:2410.12784.

Gladys Tyen, Hassan Mansoor, Victor Cărbune, Peter Chen, and Tony Mak. LLMs cannot find reasoning errors, but can correct them given the error location. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024. arXiv:2311.08516.

Peiyi Wang et al. Large language models are not fair evaluators. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024. arXiv:2305.17926.

Chenxi Whitehouse, Tianlu Wang, Ping Yu, Xian Li, Jason Weston, Ilia Kulikov, and Swarnadeep Saha. J1: Incentivizing thinking in LLM-as-a-judge via reinforcement learning. In *International Conference on Learning Representations*, 2026. arXiv:2505.10320.

Jinquan Zheng, Jia Yuan, Jiacheng Yao, Chenyang Gu, Pujun Zheng, and Guoxiu He. Mitigating selection bias in large language models via permutation-aware GRPO. *arXiv preprint arXiv:2603.21016*, 2026.

Lianmin Zheng, Wei-Lin Chiang, et al. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2023. arXiv:2306.05685.

A Data integrity and counts

Table 5 lists the counts for every cell after filtering. Of the 561 GSM8K problems, 42 had appeared in an earlier evaluation run (not in any training data). Excluding them leaves 519 problems and does not change any conclusion: skill index 0.320 (Llama), 0.249 (Mistral), 0.186 (Self_Correction), 0.487 (Qwen) (`report.json`, key `lenient_audited`). ASDiv and MATH problems do not overlap with data used in earlier phases. Bridge and report cells reuse the Phase-13/12 solutions of Self_Correction, which carry no token-cap flag ("—"); the two report cells also lack the cap flag for the `base` judge.

Table 5: Counts per cell. Cap columns: share (%) of solutions hitting the 768-token cap and of judgments hitting the judge cap (`base`, `base_c`).

Model	Dataset	Problems	$s_1 \neq s_2$	Judged	One-right	Both wrong	Sol. cap	Judge cap	Judge$_c$ cap
Llama-3.1-8B	ASDiv	822	218	430	274	156	1.6	18.1	14.2
	MATH 1–3	785	509	1021	434	587	15.7	31.9	36.5
	GSM8K	561	201	402	264	138	1.5	14.7	13.8
	SVAMP	960	304	586	389	197	2.4	17.9	12.8
	Bridge	561	240	486	288	198	—	11.2	14.4
Mistral-7B	ASDiv	822	469	933	558	375	1.8	3.5	0.3
	MATH 1–3	785	681	1377	331	1046	15.8	16.9	1.6
	GSM8K	561	398	780	411	369	2.1	3.7	0.1
	SVAMP	960	582	1159	681	478	1.4	3.0	0.1
	Bridge	561	240	486	288	198	—	11.9	1.3
Self_Correction	ASDiv	822	221	446	311	135	1.5	7.4	5.9
	MATH 1–3	785	483	970	452	518	9.0	15.0	14.4
	GSM8K	561	255	493	277	216	1.7	3.9	3.3
	SVAMP	960	246	500	371	129	1.2	3.2	2.5
	GSM8K (report)	561	240	486	288	198	—	—	3.7
	SVAMP (report)	960	246	501	367	134	—	—	3.2
Qwen2.5-7B-Instr.	ASDiv	822	178	361	256	105	0.6	9.6	1.8
	MATH 1–3	785	442	865	380	485	10.1	27.1	7.2
	GSM8K	561	174	338	241	97	0.1	23.1	5.8
	SVAMP	960	238	455	358	97	0.2	5.4	0.9
	Bridge	561	240	486	288	198	—	20.1	4.9

The *rescued* reading was checked on a seeded sample of 179 class-2 judgments (50 per model; 29 available for Self_Correction). This check was done by the assistant that ran the analysis, not by an independent annotator. Clear misreadings were 4/179 (2.2%), all on answers that are not a single number (a root, a fraction, two values). Counting ambiguous cases as errors gives 10.1% overall: 0/50 for Qwen, 10% for Llama, 12% for Mistral, and 24% for Self_Correction. Following the preregistered thresholds, the rescued reading is the main sensitivity analysis for Qwen, is reported with a caveat for Llama and Mistral, and is appendix-only for Self_Correction.

B Self_Correction deep dive

All numbers are on the original GSM8K holdout (288 one-right pairs, both orders) unless noted.

Position split. When the right solution is shown first, the untrained judge picks it 49.3% of the time (B: 34.4%, invalid 16.3%); when it is shown second, 55.2% (A: 27.8%, invalid 17.0%). Skill index +0.215 [0.128, 0.304], single-order accuracy 52.3%. On SVAMP: skill +0.248 [0.166, 0.332].

Training the judge. Order-swapped DPO (p12) raises tie-break from 0.601 to 0.653 (+5.2 points [1.6, 8.8], Holm $p = 0.016$); DPO on the constrained judge (p13b) beats the constrained baseline by 8.3 points (Holm $p = 0.0008$) and cuts invalid choices from 4.9% to 2.1%. Consistent-only

accuracy (71–79%) overstates quality: over all one-right pairs the best judge is right on 41–47% and inconsistent on about half.

Is the judge better than random? Choosing between s_1 and s_2 at random is no better than keeping s_1 (65.7 vs 66.1% on GSM8K). The trained judge adds 4.2 points over random choice on GSM8K and 3.1 on SVAMP; a self-check that falls back to a third sample reaches vote@3 (72.2 vs 72.0%) but not vote@5 (77.5%).

Hidden states. With one protocol for all four judges, a linear probe on the judge-prompt state reaches 65–69% across datasets (SVAMP→GSM8K) and ≈71.5% within GSM8K, with no effect of judge training. Surface features alone reach AUC 0.742; hidden + surface 0.759, and the probe is right on 64% of the rows where surface features fail, so the two signals are only partly distinct. The probe is not order-invariant (about 44% of pairs flip). Mean token log-probability selects the right solution 56–61% of the time; agreement with one extra greedy solve reaches 81% but costs a solver call. A logit lens on the verdict letter becomes informative only from layer ≈22, several layers after the linear probe.

Origin of the second-position preference. Every row of the self-correction SFT format is a failed attempt followed by the verified correction, so the data carry a "later answer is right" prior by construction; this is *consistent with* the preference but does not show that it caused it. Renaming the labels (first shown called "Solution B") leaves the preference unchanged, so it follows position, not the label. Restating the problem after the solutions removes the net preference (+0.090 → +0.009) and raises coverage, but not accuracy (nominal $p = 0.04$, not significant after Holm).

C Bridge set and parent-model checks

Four judges re-judged the same 288 GSM8K pairs written by Self_Correction (bridge set). Tie-break: Qwen-Instruct 0.774, Llama 0.729, Mistral 0.595, Self_Correction itself 0.601. Qwen-Instruct shows no net position preference here (−0.024 [−0.073, 0.026]). This gap is *not* evidence that SFT harmed arbitration. On its own conflicting solutions the parent reaches only 0.61–0.66, it prefers the first position, and the two sets of pairs differ in difficulty. A stronger judge does well on a weaker model's disagreements.

D Truncation labels and token-cap sensitivity

Every output that hit its token cap was labelled automatically, in a fixed order: (c) off-topic, garbage or a language switch; (b) looping (a 10-word sequence or a sentence repeated verbatim ≥3 times, or a templated list); (d) for judgments only, abandoning both solutions and re-solving; (a) otherwise, still on track. Loops dominate the capped judgments of Llama (65–91% of its capped `base` judgments in the four grid datasets) and Self_Correction (73–100% in the pilot). Mistral and Qwen also re-solve or are cut while still on track. Raising Llama's cap to 1536 tokens lowers the capped share (18.1% → 12.8% on ASDiv, 31.9% → 15.8% on MATH), but 74–76% of the outputs still capped are loops, and the single-order gap does not shrink (Section 4.2). Full tables: `paper/results/phase14_truncation.md`.

E Tie-break identity

Per pair, tie-break minus mean single-order accuracy equals

$$\tfrac{1}{2}\,\mathbf{1}\big[\text{no order picks the right solution and one order gives a third answer}\big];$$

a judge with pure position bias and no invalid choices gets 0.

F Prompts, hyperparameters, code and hashes

See `phase14/DATA_CATALOG.md` and `phase14/DATA_MANIFEST.json` in the repository.